

Arrays

LEVEL 1: Array Basics

Program 1: Input and Display Array Write a program that takes N numbers as input, stores them in an array, and displays all elements in a single line.

Program 2: Display Array in Reverse Order Write a program that inputs an array and displays elements in reverse order (without modifying the array).

Program 3: Sum of Array Elements Write a program that calculates and displays the sum of all elements in an array.

Program 4: Average of Array Elements Write a program that calculates the average of all array elements and displays it.

Program 5: Count Elements Greater Than X Write a program that counts how many elements in the array are greater than a given number X.

Program 6: Count Elements Less Than X Write a program that counts how many elements are less than a given number X.

Program 7: Count Even and Odd Numbers Write a program that counts how many even and odd numbers are present in the array separately.

Program 8: Count Positive, Negative, and Zero Write a program that counts positive, negative, and zero elements separately in the array.

Program 9: Sum of Even Numbers Only Write a program that calculates the sum of only even numbers in the array.

Program 10: Sum of Odd Numbers Only Write a program that calculates the sum of only odd numbers in the array.

Program 11: Product of All Elements Write a program that calculates the product of all elements in the array.

Program 12: Check if Element Exists Write a program that takes a number X as input and checks if it exists in the array. Display "Found" or "Not Found".

LEVEL 2: Finding & Searching

Program 13: Find Maximum Element Write a program that finds and displays the largest element in the array.

Program 14: Find Minimum Element Write a program that finds and displays the smallest element in the array.

Program 15: Find Maximum and Minimum Together Write a program that finds both maximum and minimum elements in a single traversal and displays both.

Program 16: Find Position of Maximum Write a program that finds the index/position where the maximum element is located.

Program 17: Find Position of Minimum Write a program that finds the index/position where the minimum element is located.

Program 18: Linear Search Write a program that searches for an element X and returns its first position. Display -1 if not found.

Program 19: Count Occurrences of Element Write a program that counts how many times a specific element X appears in the array.

Program 20: Find First Occurrence Write a program that finds the first position where element X appears. Display -1 if not found.

Program 21: Find Last Occurrence Write a program that finds the last position where element X appears. Display -1 if not found.

Program 22: Find All Positions of Element Write a program that displays all positions where element X appears in the array.

Program 23: Check for Duplicates Write a program that checks if any element appears more than once in the array. Display "Yes" or "No".

Program 24: Find First Duplicate Write a program that finds the first element that appears more than once in the array.

Program 25: Count Distinct Elements Write a program that counts how many unique/distinct elements are in the array.

Program 26: Frequency of Each Element Write a program that displays the frequency of each unique element in the array.

Program 27: Find Element with Maximum Frequency Write a program that finds which element appears most frequently in the array.

LEVEL 3: Array Manipulation - Basic Operations

Program 28: Reverse Array (In-Place) Write a program that reverses the array by modifying it. Example: [1,2,3,4,5] becomes [5,4,3,2,1].

Program 29: Swap First and Last Element Write a program that swaps the first and last elements of the array.

Program 30: Swap Elements at Two Positions Write a program that swaps elements at positions i and j (user input).

Program 31: Rotate Array Left by 1 Write a program that shifts all elements one position to the left (first goes to end). Example: [1,2,3,4,5] → [2,3,4,5,1].

Program 32: Rotate Array Right by 1 Write a program that shifts all elements one position to the right (last comes to front). Example: [1,2,3,4,5] → [5,1,2,3,4].

Program 33: Rotate Array Left by K Positions Write a program that rotates the array left by K positions. Example: [1,2,3,4,5], K=2 → [3,4,5,1,2].

Program 34: Rotate Array Right by K Positions Write a program that rotates the array right by K positions. Example: [1,2,3,4,5], K=2 → [4,5,1,2,3].

Program 35: Copy Array to Another Array Write a program that copies all elements from one array to another array.

Program 36: Insert Element at Specific Position Write a program that inserts a new element at position i and shifts remaining elements right.

Program 37: Delete Element at Specific Position Write a program that deletes element at position i and shifts remaining elements left.

Program 38: Insert Element in Sorted Array Write a program that inserts element in correct position to maintain sorted order.

Program 39: Remove All Occurrences of X Write a program that removes all occurrences of element X from the array.

Program 40: Replace All X with Y Write a program that replaces all occurrences of X with Y in the array.

Program 41: Merge Two Arrays Write a program that merges two arrays into a third array.

Program 42: Concatenate Two Arrays Write a program that concatenates array B at the end of array A.

Program 43: Split Array into Two Equal Parts Write a program that splits array into two equal halves (if size is even).

Program 44: Separate Even and Odd Elements Write a program that rearranges array with all even numbers first, then odd numbers. Example: [1,2,3,4,5] → [2,4,1,3,5].

Program 45: Move All Zeros to End Write a program that moves all zeros to the end while maintaining relative order of non-zero elements. Example: [0,1,0,3,12] → [1,3,12,0,0].

LEVEL 4: Array Analysis & Properties

Program 46: Check if Array is Sorted (Ascending) Write a program that checks if the array is sorted in strictly ascending order. Display "Yes" or "No".

Program 47: Check if Array is Sorted (Descending) Write a program that checks if the array is sorted in strictly descending order.

Program 48: Check if Array is Sorted (Non-Decreasing) Write a program that checks if array is sorted in non-decreasing order (allows equal consecutive elements).

Program 49: Check if Array is Palindrome Write a program that checks if array reads the same forwards and backwards. Example: [1,2,3,2,1] → Yes.

Program 50: Sum at Even Indices Write a program that calculates sum of elements at even indices (0, 2, 4, ...).

Program 51: Sum at Odd Indices Write a program that calculates sum of elements at odd indices (1, 3, 5, ...).

Program 52: Difference Between Even and Odd Index Sums Write a program that finds difference between sum at even indices and sum at odd indices.

Program 53: Find Range (Max - Min) Write a program that finds the range of array elements (difference between max and min).

Program 54: Find Second Largest Element Write a program that finds the second largest element without sorting the array.

Program 55: Find Second Smallest Element Write a program that finds the second smallest element without sorting.

Program 56: Find Kth Largest Element Write a program that finds the Kth largest element (K given by user).

Program 57: Find Kth Smallest Element Write a program that finds the Kth smallest element.

Program 58: Count Elements Greater Than Average Write a program that counts how many elements are greater than the array average.

Program 59: Count Elements Less Than Average Write a program that counts how many elements are less than the array average.

Program 60: Find Longest Increasing Sequence Length Write a program that finds length of longest consecutive increasing sequence.
Example: [1,2,3,1,2,3,4] → Length: 4.